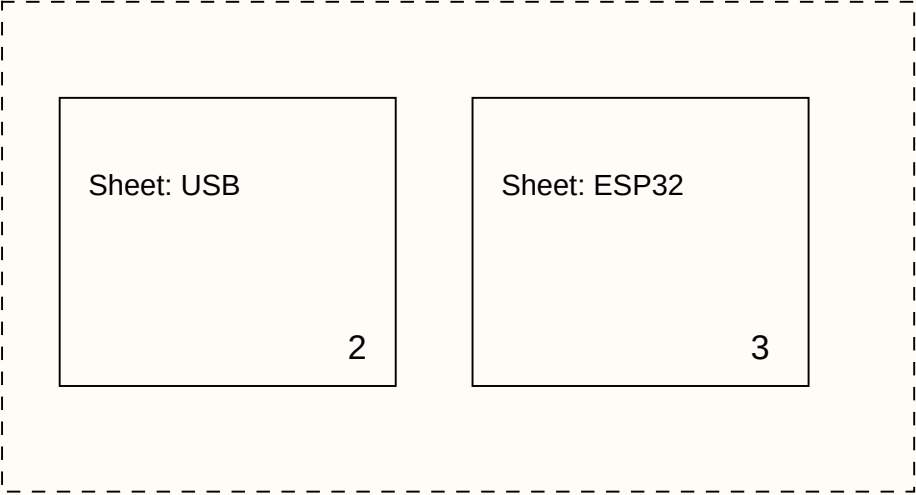
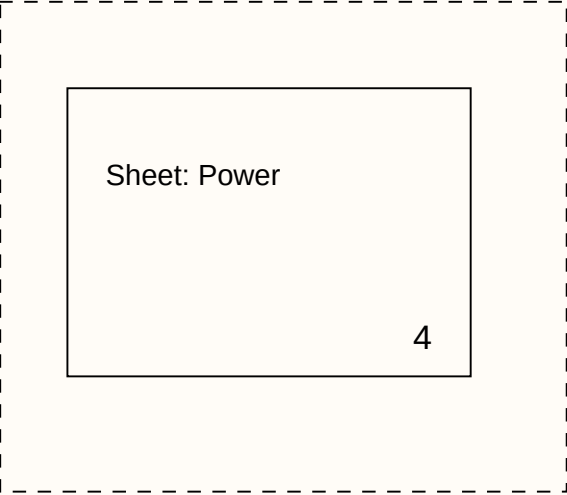


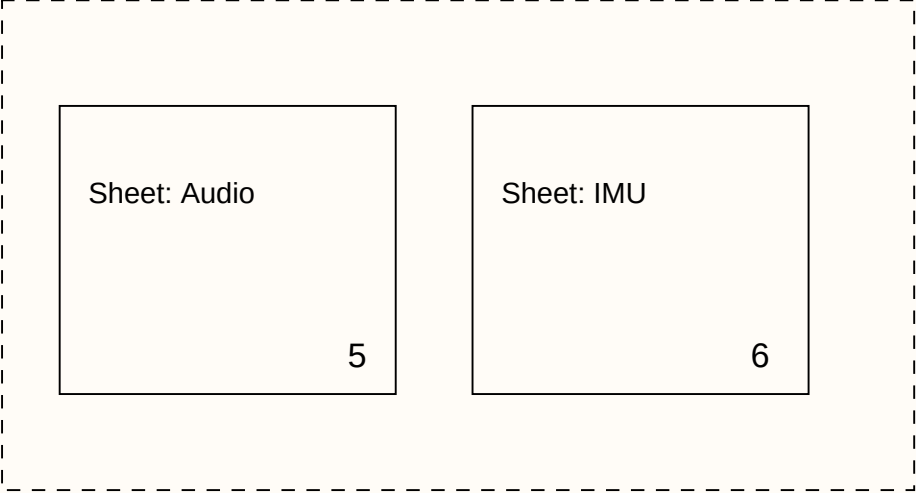
Microcontroller



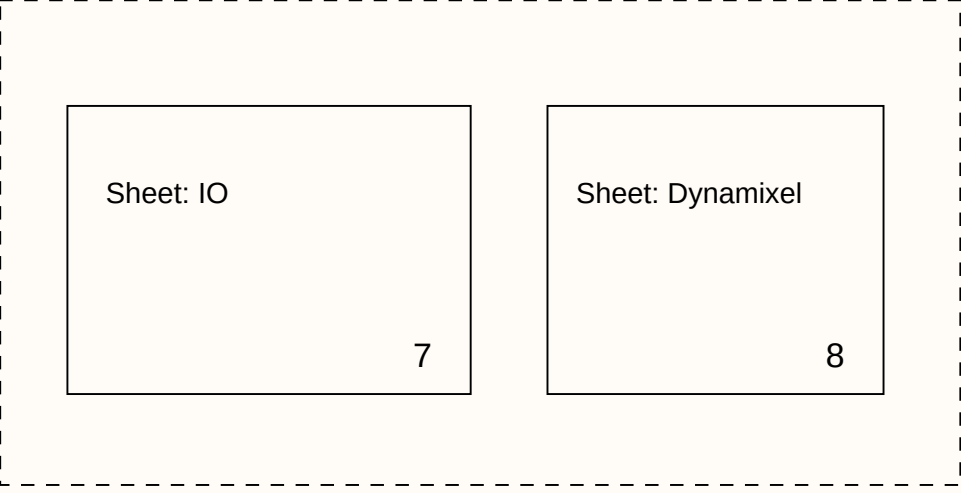
Power



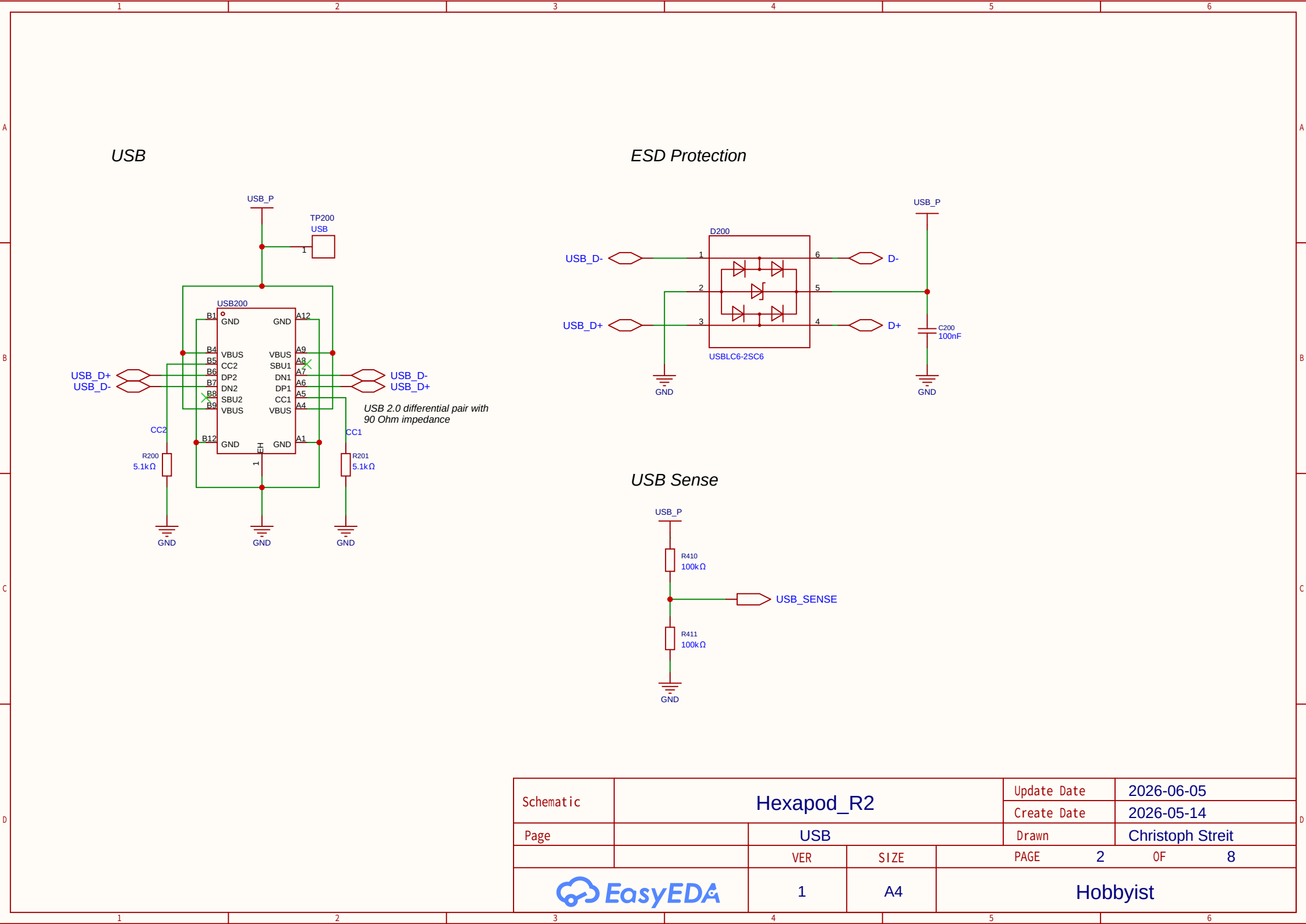
Audio & IMU



I/O



Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-04-06
Page		Overview		Drawn	Christoph Streit
		VER	SIZE	PAGE	1 OF 8
EasyEDA		1	A4	Hobbyist	



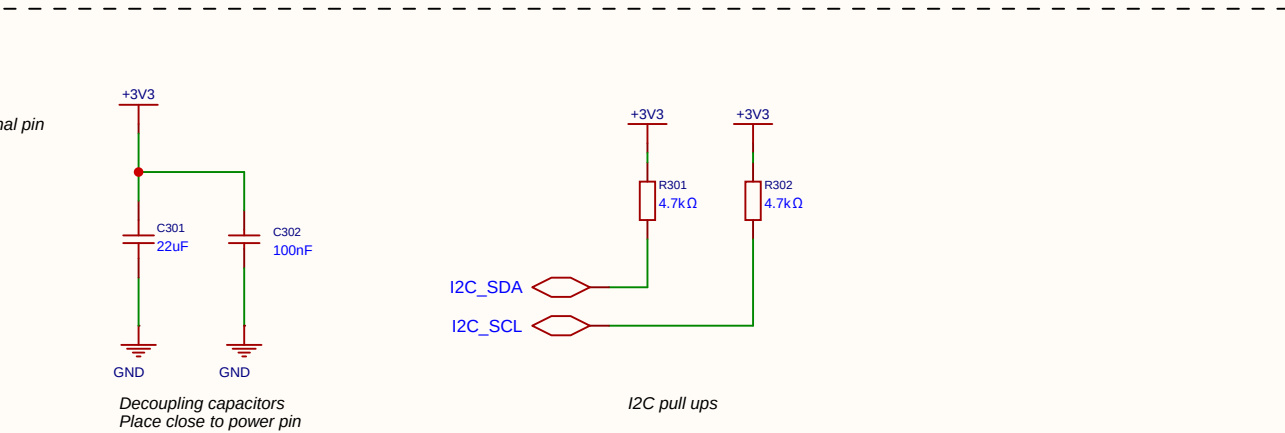
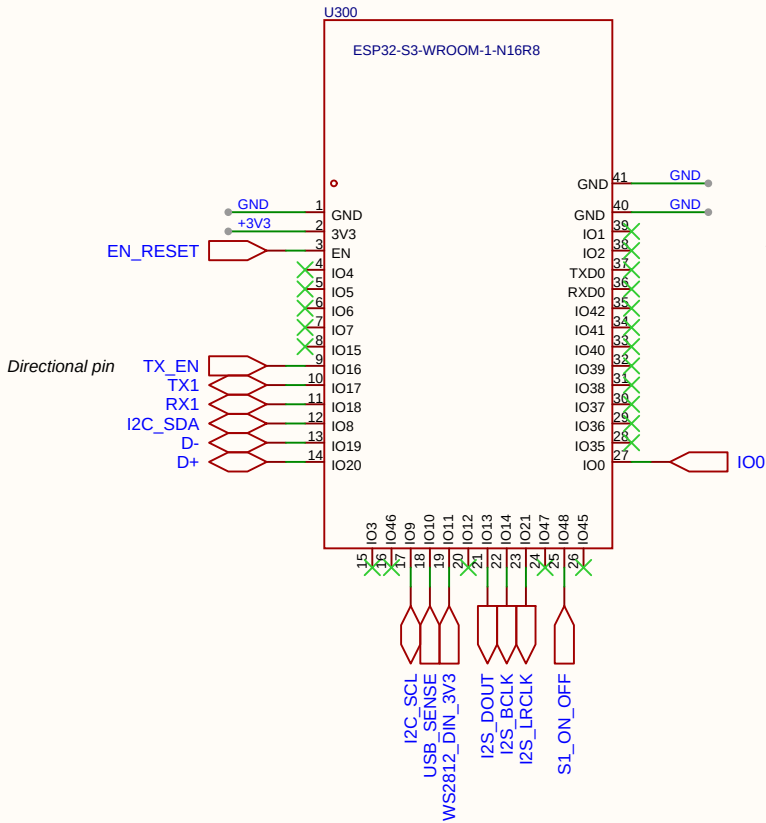
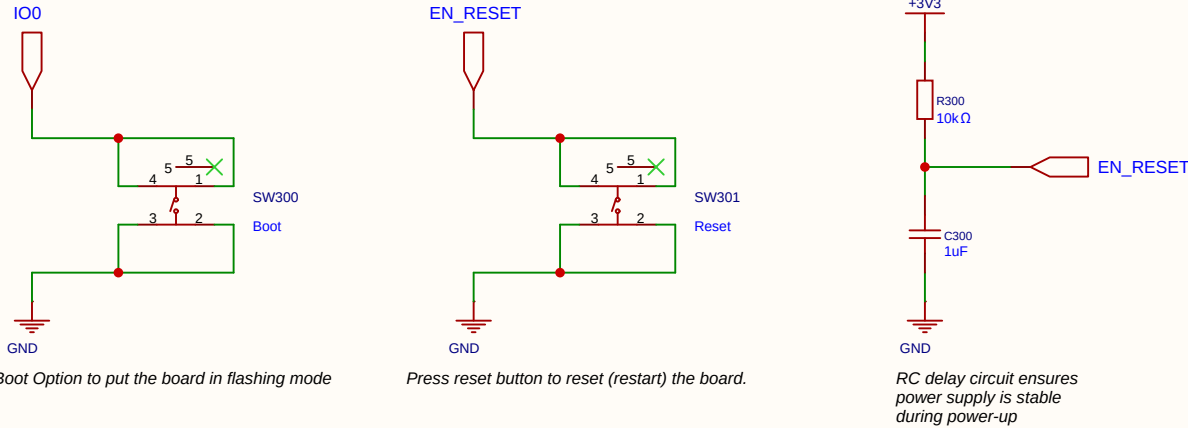
ESP32--S3-WROOM-1-N16R8

Datasheet: https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

Note:
Ensure that ESP32 strapping pins do not conflict with boot sequence IO0, IO3, IO45, IO46.

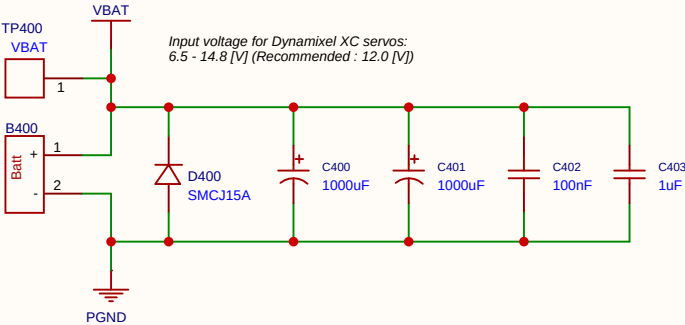
Pins IO35, IO36, and IO37 are connected to the Octal SPI PSRAM and are not available for other uses.

Boot Option and Reset Button



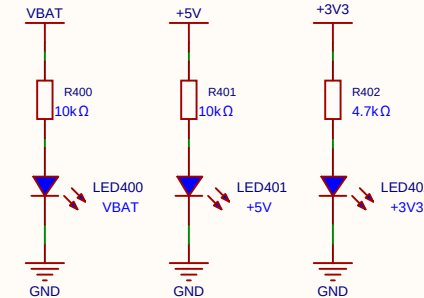
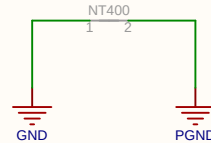
Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-05-16
Page	ESP32			Drawn	Christoph Streit
		VER	SIZE	PAGE	3 OF 8
EasyEDA		1	A4	Hobbyist	

Battery/LIPO

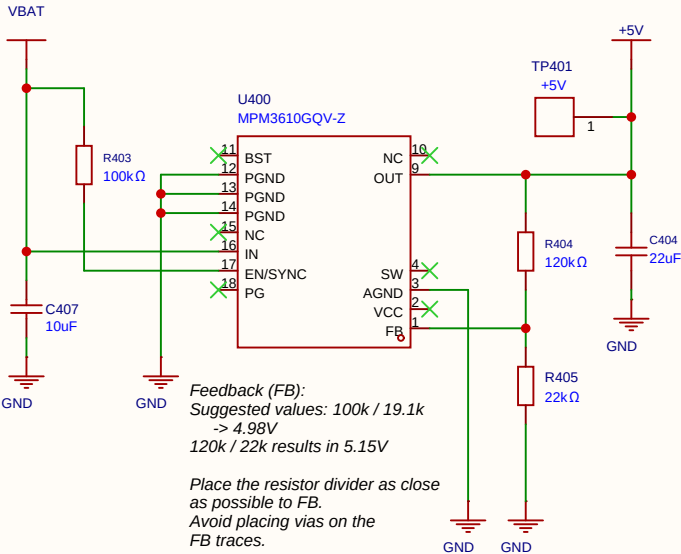


Net Tie and Indicator LEDs

Star grounding is used to connect LIPO ground and logic ground to a single, central point. This prevents ground loops and reduces electrical noise by ensuring all components share a common, stable reference potential.

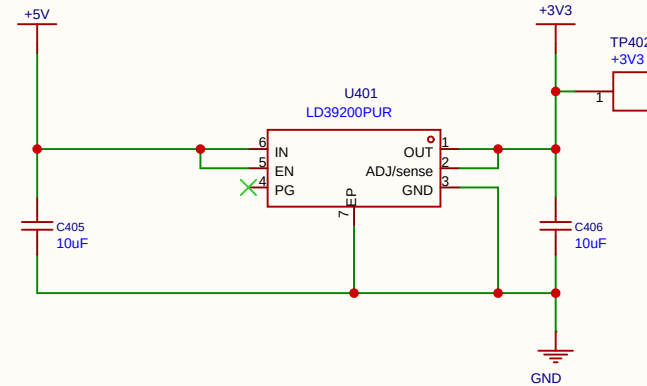


+5V Step-Down Converter



+3.3V LDO

3.3V to power the ESP, the display and other devices

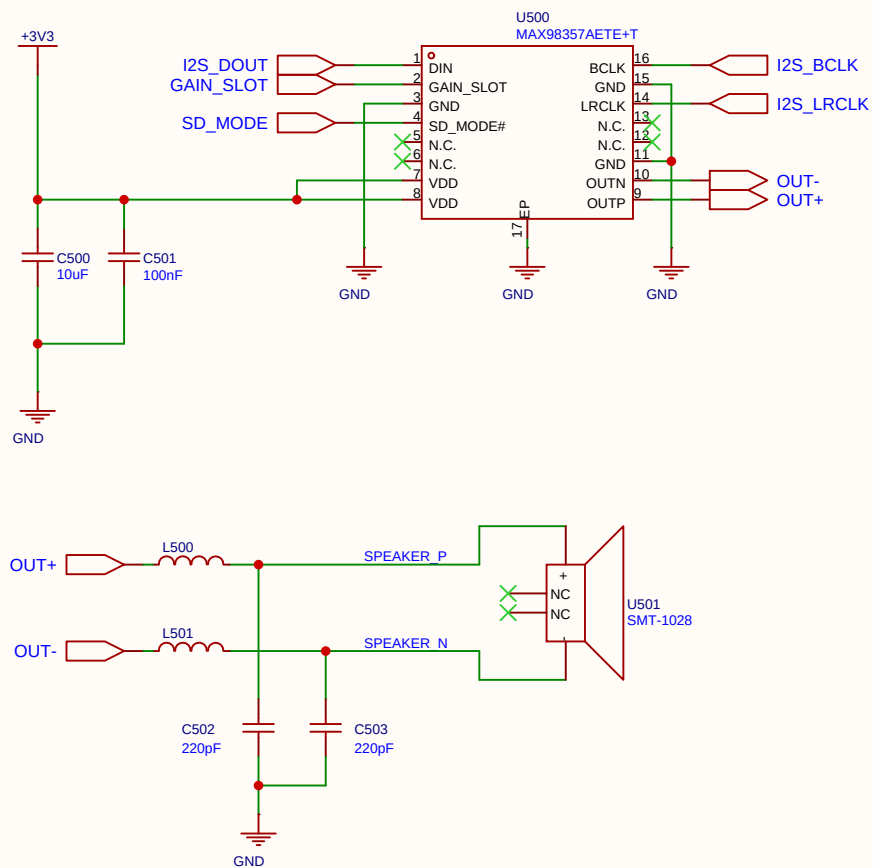


Datasheet: <https://www.st.com/resource/en/datasheet/ld39200.pdf>

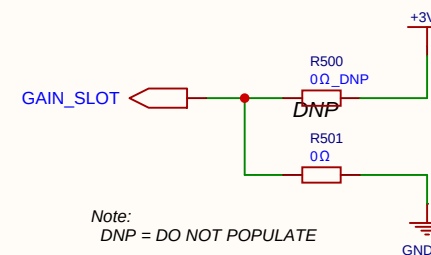
Datasheet: <https://tinyurl.com/bdhxaekf>

Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-04-29
Page		Power		Drawn	Christoph Streit
		VER	SIZE	PAGE	4 OF 8
EasyEDA		1	A4	Hobbyist	

Amplifier



Gain Selection	
GAIN_SLOT	GAIN
Connect to GND through 100kΩ resistor	+15 dB
Connect to GND	+12 dB
Unconnected	+9 dB
Connect to VDD	+6 dB
Connect to VDD through 100kΩ resistor	+3 dB

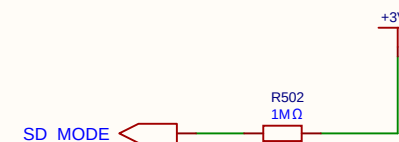


Channel Selection (SD_MODE)

LEFT = VDD (0 Ω)

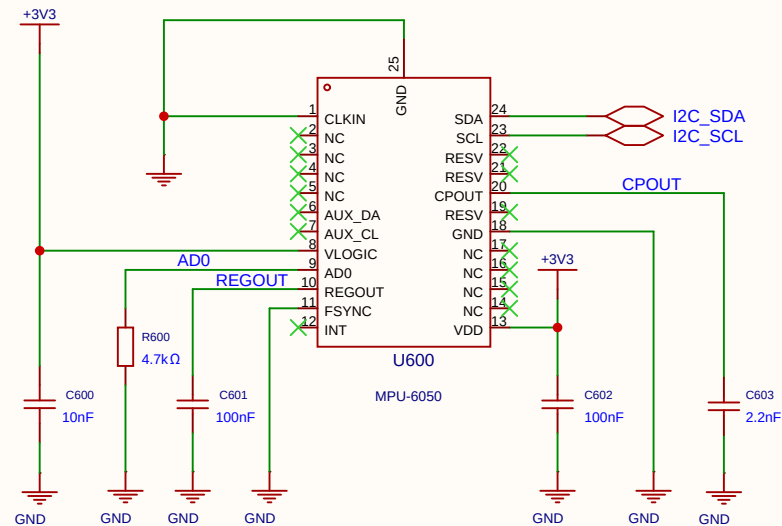
RIGHT = 370K
= $94 \times 5 - 100$ (RSMALL (kΩ) = $94.0 \times VDDIO - 100$)

LEFT/2 + RIGHT/2 = 1.011M
= $222.2 \times 5 - 100$ (RLARGE (kΩ) = $222.2 \times VDDIO - 100$)



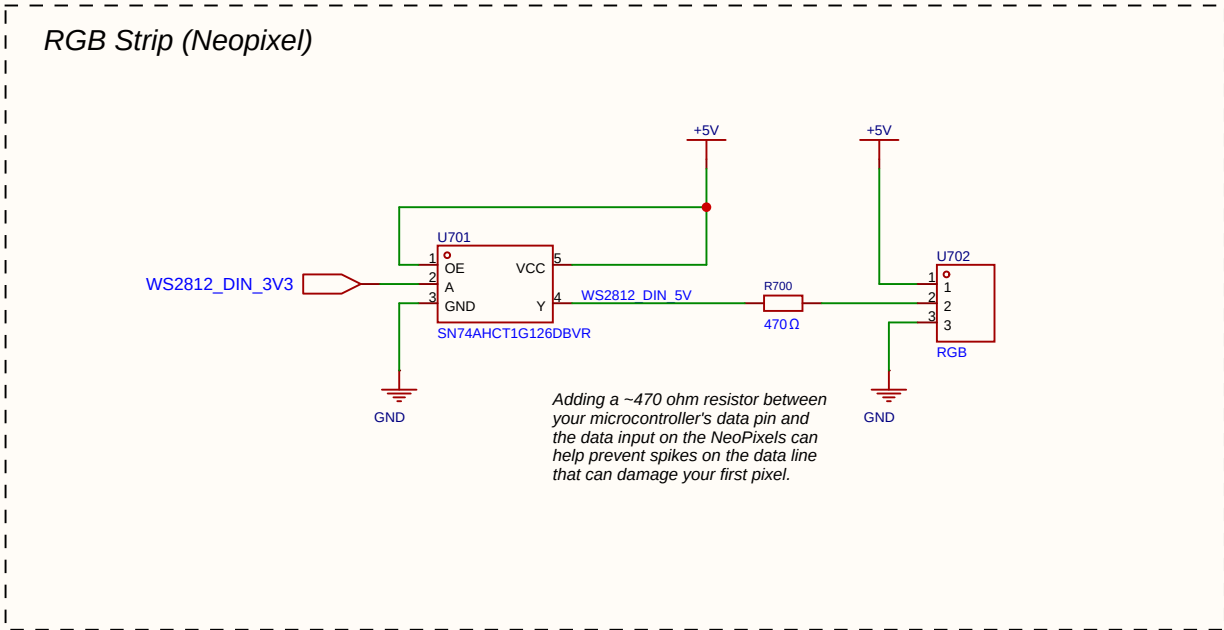
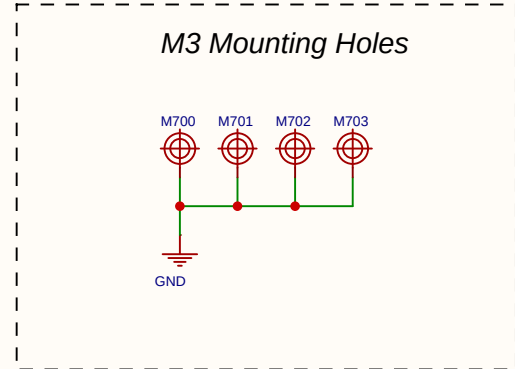
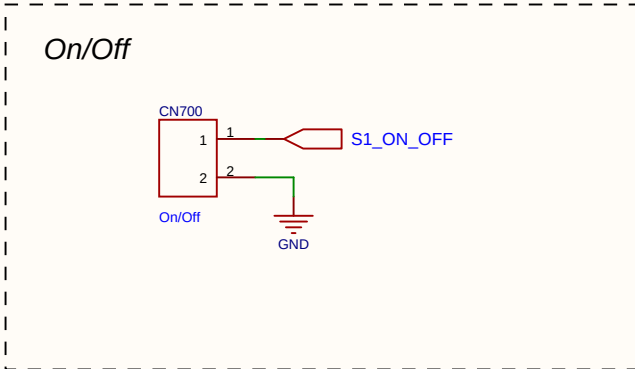
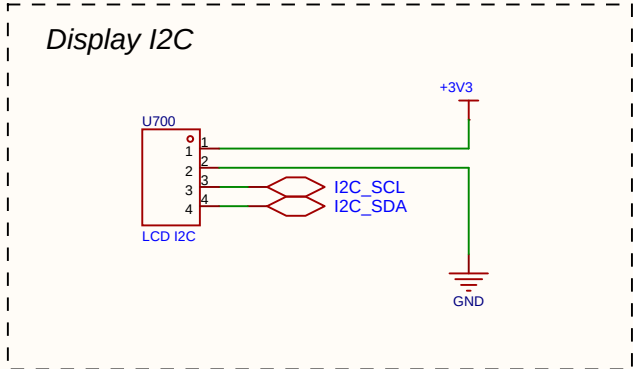
Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-04-10
Page		Audio		Drawn	Christoph Streit
		VER	SIZE	PAGE	5 OF 8
EasyEDA		1	A4	Hobbyist	

IMU

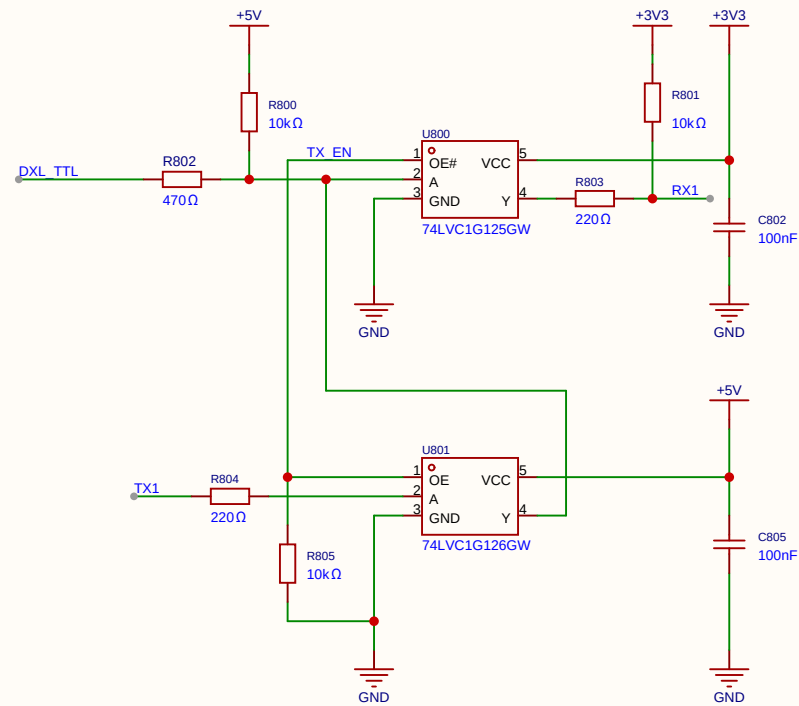
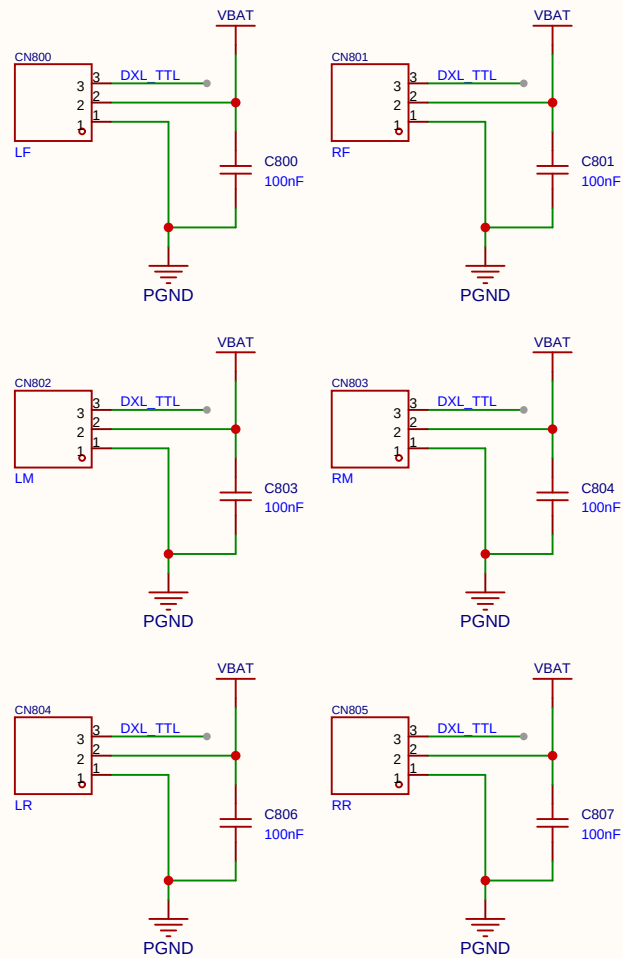


Datasheet: https://product.tdk.com/system/files/dam/doc/product/sensor/mortion-inertial/imu/data_sheet/mpu-6000-datasheet1.pdf

Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-04-12
Page		IMU		Drawn	Christoph Streit
		VER	SIZE	PAGE	6 OF 8
EasyEDA		1	A4	Hobbyist	



Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-05-15
Page		I/O		Drawn	Christoph Streit
		VER	SIZE	PAGE	7 OF 8
EasyEDA		1	A4	Hobbyist	



Schematic	Hexapod_R2			Update Date	2026-06-05
				Create Date	2026-05-16
Page		Dynamixel		Drawn	Christoph Streit
		VER	SIZE	PAGE	8 OF 8
EasyEDA		1	A4	Hobbyist	